AIML TRAINING PROGRAM • FINAL PROJECT PRESENTATION

Customer Segmentation, Customer Type Prediction & Country Growth Prediction

Turning a year of online retail transactions into decisions about marketing, retention, and where to grow next.

TEAM

Sanvi • Shivansh • Shayan • Sanskar

Batch: CSE — AIML Training Program

27 July 2026

AGENDA

What We'll Cover

01 **Training Program Recap**
The 10-day journey from Python basics to Machine Learning

02 **Project Overview**
The business problem and the three questions we're answering

03 **Methodology**
Data cleaning, feature engineering, and model building

04 **Results**
Customer segments, predictions, and growth markets

05 **Business Impact**
How this turns into marketing, retention and expansion decisions

Building the Foundations

Learned core Python syntax and data structures, then applied object-oriented programming to model real-world entities.

Day 1

Python Fundamentals

- Variables & data types
- Operators & conditions
- Loops, calculator & grade programs

Day 2

Functions & Collections

- Functions
- Lists, tuples, sets, dictionaries
- File handling

Day 3

OOP & NumPy

- Classes & objects
- Inheritance
- NumPy arrays & operations

PRACTICE SNIPPET — OOP with Inheritance

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age
class Student(Person):
    def __init__(self, name, age, roll_no, marks):
        super().__init__(name, age)
        self.roll_no = roll_no
        self.marks = marks
```

Working With Data & Seeing It Clearly

Used Pandas to clean and analyze tabular data, then visualized patterns and trends with Matplotlib chart types.

Day 4

Pandas

- DataFrames & CSV files
- Data cleaning & filtering
- Grouping & analysis

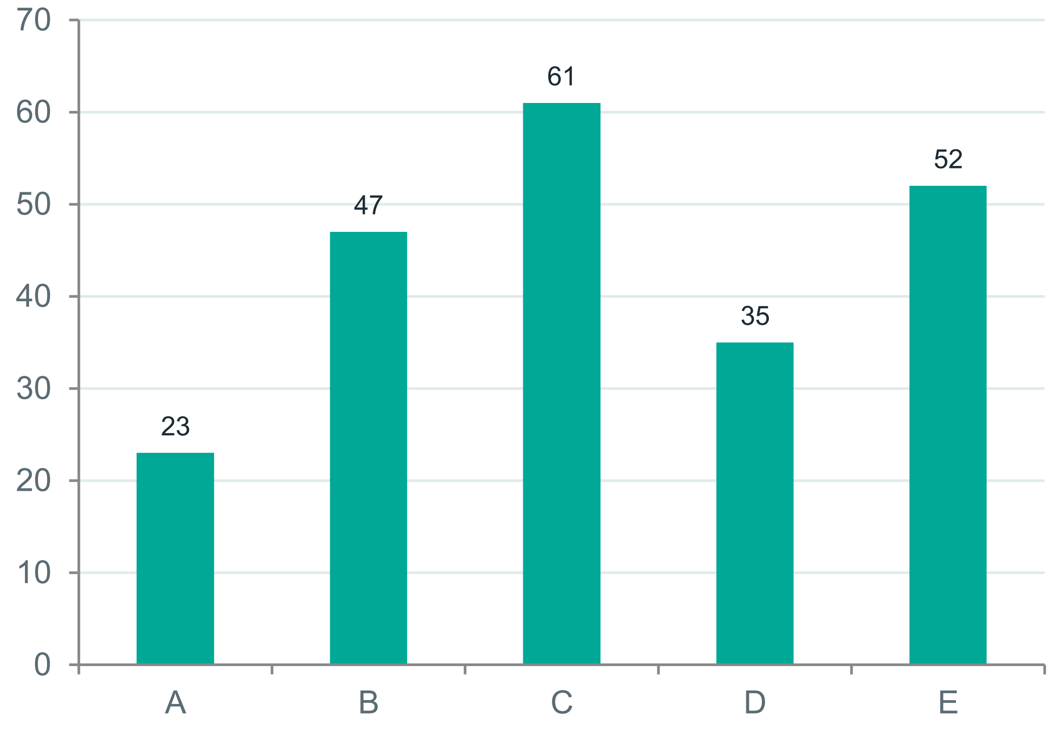
Day 5

Visualization

Matplotlib chart types practiced:

Bar Chart | Pie Chart | Histogram | Scatter Chart | Heat Map

PRACTICE OUTPUT — Bar Chart Example



From ML Concepts to Evaluation

Built regression and classification models, then learned to measure their performance using standard evaluation metrics.

Day 6

ML Introduction

- AI vs ML, workflow, datasets
- Supervised vs unsupervised learning
- Reinforcement learning (definition only)

Day 7

Regression

- Linear & Logistic Regression
- Decision Trees
- Prediction projects

Day 8

Classification

- KNN, Decision Tree
- Random Forest (intro)
- K-Means clustering

Day 9

Evaluation

- Accuracy, Precision, Recall, F1
- Feature scaling
- ML pipeline (definitions)

Note: Day 9 (Evaluation) covered definitions only — no hands-on scoring exercises.

PROJECT OVERVIEW

From Raw Transactions to Business Decisions

A full year of an online retail company's transaction records — every item bought, by whom, when, and from which country — analyzed with machine learning to answer three business questions.

Q1

Who are our customers?

Group ~4,000+ customers into natural segments based on real shopping behaviour — using KMeans clustering.

Q2

What type is a new customer?

Instantly classify a new customer's value tier from a few transactions — using a Random Forest model.

Q3

Where should we grow?

Identify which countries' sales are accelerating fastest — using Linear Regression on monthly revenue trends.

Dataset Snapshot

541,909	397,884	4,338	37
Raw Transaction Rows	Rows After Cleaning	Unique Customers	Countries

Data period: 2010-12-01 → 2011-12-09

Cleaning steps applied:

- Removed rows with missing CustomerID (can't be tied to a customer)
- Removed cancelled orders (InvoiceNo starting with "C")
- Removed rows with non-positive Quantity or UnitPrice (returns / data errors)
- Added a TotalPrice column = Quantity × UnitPrice

Building a Customer-Level Profile

Every transaction row was rolled up into one profile per customer, with four behavioural features:

`TotalSpent`

How much they spend in total

`TotalQuantity`

How many items they buy

`TotalTransactions`

How often they place orders

`AveragePrice`

What price point they buy at

Outlier Removal

Customers above the 99th percentile on any feature were removed to prevent a few extreme accounts from distorting the clusters.

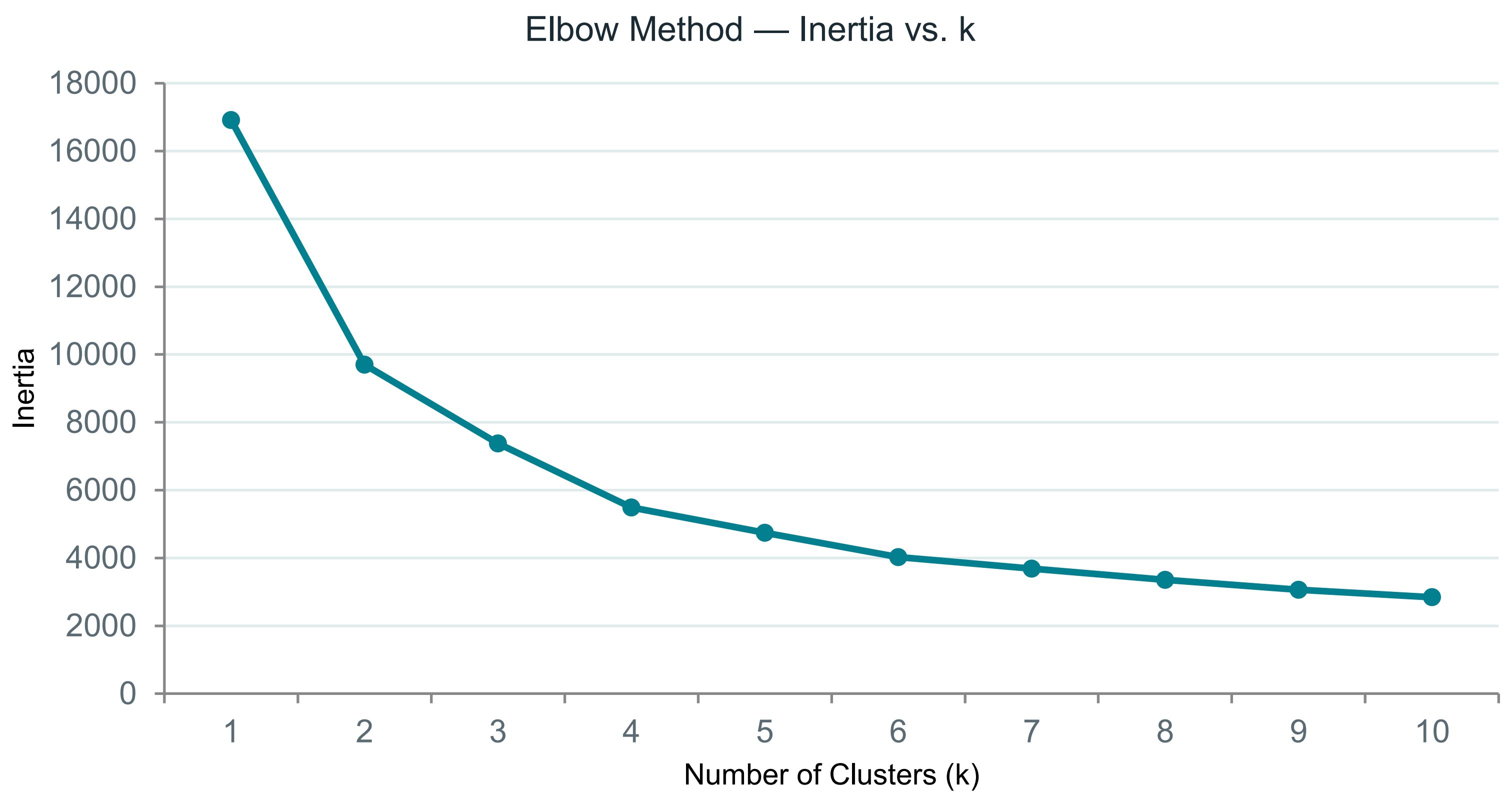
4,338 → 4,228 customers

Feature Scaling

All four features were standardized (StandardScaler) so no single feature — like TotalSpent — dominates the clustering just because it's on a bigger scale.

Choosing the Right Number of Clusters

We tested k = 1 to 10 and plotted inertia against k. The curve flattens noticeably at k = 4 — and 4 also gives clean, business-friendly segments.



CHOSEN VALUE

k = 4

Maps naturally onto four business-friendly segments:

- **Low Value**
- **Regular**
- **High Value**
- **Premium**

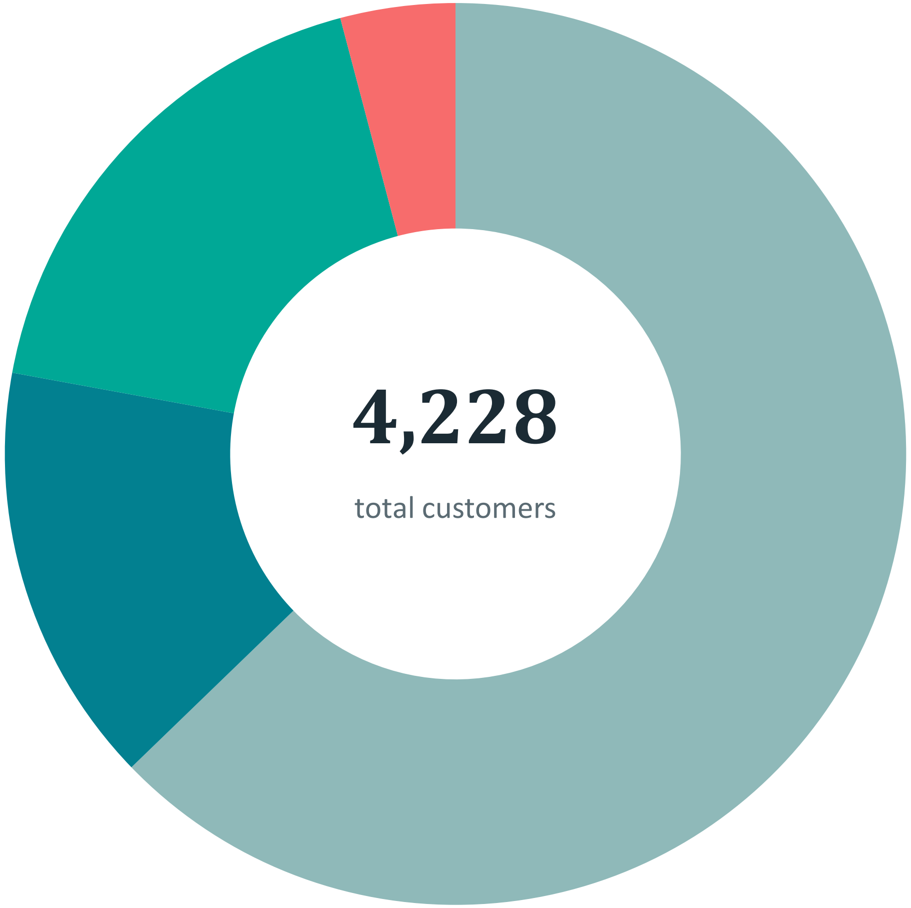
Four Natural Customer Segments

AVERAGE PROFILE PER SEGMENT

Segment	Spend	Orders
Premium Customer	£8,308.1	15.83
High Value Customer	£2,914.25	7.59
Regular Customer	£674.74	2.2
Low Value Customer	£613.34	2.13

Clusters are ranked by average spend into human-readable customer types — Low Value, Regular, High Value, and Premium.

How the Customer Base Breaks Down



Low Value Customer
2,654 customers
62.8%

Regular Customer
639 customers
15.1%

High Value Customer
761 customers
18.0%

Premium Customer
174 customers
4.1%

Predicting a New Customer's Type Instantly

KMeans alone can only assign a new point to its nearest centroid. So the cluster labels become ground-truth, and a Random Forest classifier is trained on them — giving a real model that predicts customer type directly for new data.

Data was split manually — 80% training, 20% testing — using NumPy to shuffle row indices.

3,382

Training Samples

846

Test Samples

TEST ACCURACY

98.58%

Random Forest, 200 trees, evaluated on the held-out 20% test set

CONFUSION MATRIX (TEST SET)

Actual \ Pred	Prem	Low	Reg	High
Prem	**37**	0	0	3
Low	0	**518**	0	1
Reg	0	0	**127**	1
High	4	3	0	**152**

Prem = Premium, Reg = Regular, High = High Value

Try It: Classifying New Customers

Feed the model a few raw numbers, and it instantly returns a customer type — no need to re-run the clustering.

CUSTOMER 1

Total Spent	**£150**
Total Quantity	**60**
Transactions	**1**
Avg. Price	**£2.5**

PREDICTED TYPE

Low Value Customer

CUSTOMER 2

Total Spent	**£1800**
Total Quantity	**900**
Transactions	**6**
Avg. Price	**£3**

PREDICTED TYPE

High Value Customer

CUSTOMER 3

Total Spent	**£7000**
Total Quantity	**4200**
Transactions	**15**
Avg. Price	**£2.9**

PREDICTED TYPE

Premium Customer

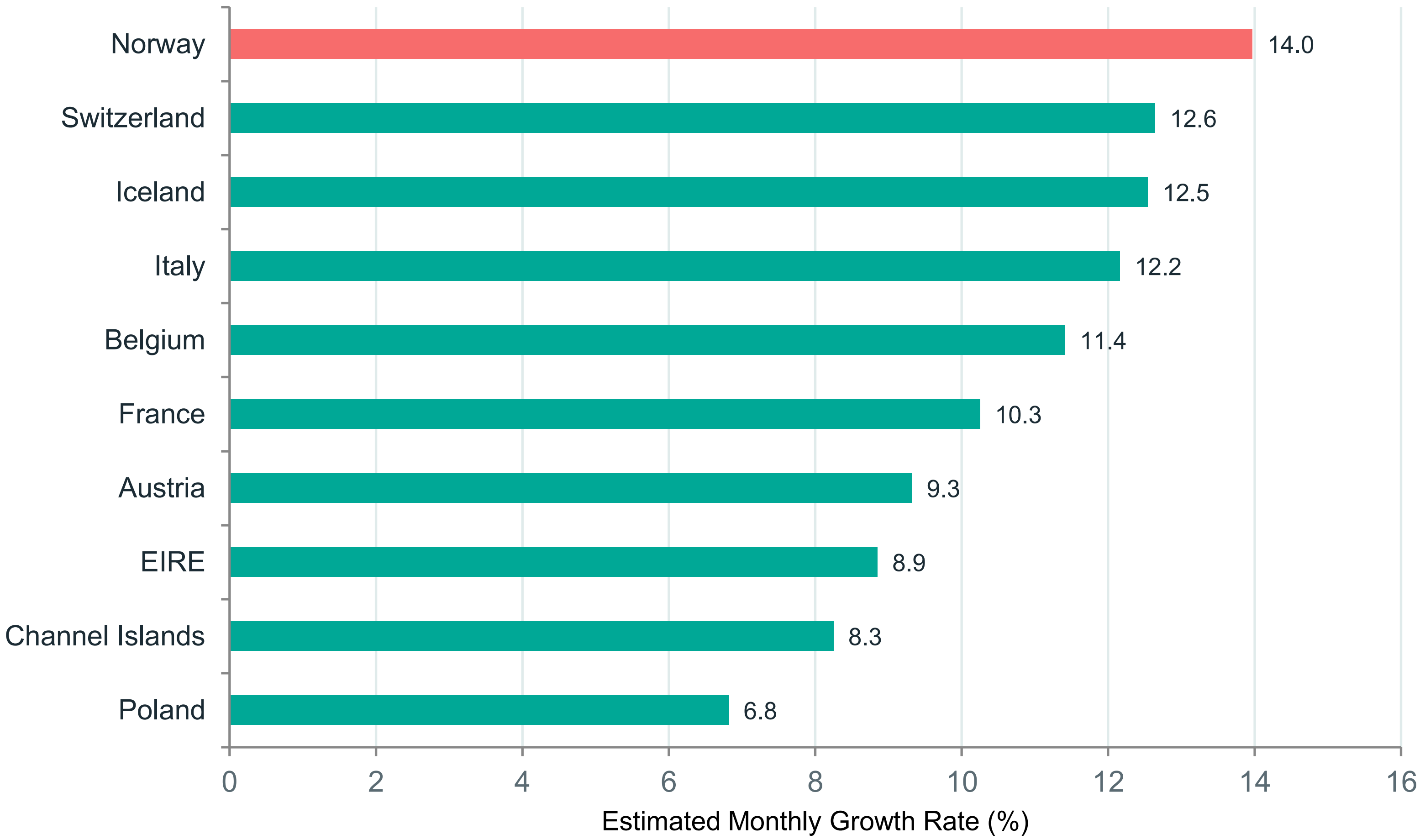
Finding the Fastest-Growing Markets

Using the Country and InvoiceDate columns (not used in segmentation), we model monthly revenue trends per country with Linear Regression.

1. Aggregate revenue by country × month
2. Drop the last month (partial data, would look like a fake decline)
3. Keep countries with ≥6 months of history, exclude UK (≈90% of revenue) — 20 countries qualify
4. Fit Linear Regression: revenue vs. month index; slope → % growth rate

Growth rate is expressed as a % (slope ÷ average monthly revenue), so small and large countries can be compared fairly.

Top 10 Countries by Predicted Sales Growth



#1 FASTEST-GROWING MARKET

Norway

~14%

estimated revenue growth per month

Average monthly revenue: £3,708.86

Switzerland and Iceland follow closely behind as the next fastest-growing markets.

Revenue Trend — Top 3 Growing Countries

BUSINESS IMPACT

Turning Analysis Into Action

Targeted Marketing

Send premium offers and loyalty perks to Premium/High Value customers instead of one email to everyone.

Retention Priority

Focus retention effort on Regular/High Value customers who are worth saving, not spread evenly.

Personalized Pricing

AveragePrice reveals frequent-cheap vs. rare-expensive buyers — tailor bundles accordingly.

Instant Classification

New customers get a value tier from just a few early transactions — no months of waiting.

Targeted Expansion

Put marketing spend and local promotions into Norway, Switzerland, and Iceland first.

Revenue Efficiency

Resources go where they generate the most return — growth without necessarily increasing spend.

IN ONE LINE

Unsupervised learning discovers customer segments, supervised learning classifies new customers instantly, and regression identifies the fastest-growing markets — turning raw transactions into decisions about marketing, retention, and expansion.

Thank You

Sanvi • Shivansh • Shayan • Sanskar

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